Guardian AI Analyst Manual: Deep Dive Guide for SOC Team

1. Introduction: What is the Guardian AI Analyst?

Welcome to the **Guardian AI Analyst** (powered by Gemini)! This is not just a search tool; it's an advanced security assistant designed to drastically cut down your response time.

1.1. What is its Core Role? (The 3 Pillars of AI)

The AI Analyst is built on three fundamental pillars of modern security operations:

Pillar	Detailed Explanation	Core Function
1. Threat Intelligence Fusion	The AI pulls information from various sources (like AbuseIPDB, GeoIP) and combines it with your local logs *in real-time*. This helps instantly determine if an IP is malicious or if an event originates from a high-risk country.	**Enrichment:** Adding external context to local logs.
2. Correlation & Prioritization	It constantly scans for suspicious *sequences* of events (e.g., three failed logins followed by one success) and prioritizes them based on a risk register (R-IDs).	**Focus:** Identifying high-fidelity threats that need human attention immediately.
3. SOAR and Remediation	For high-severity alerts, it suggests and can sometimes execute immediate containment actions (e.g., blocking an IP via firewall rules, if admin privileges are available).	**Response:** Automating the first steps of incident handling.

2. Core Concepts Explained Simply

Term	Simple Meaning	Why it Matters for You
SIEM (Security Information and Event Management)	This is the *central brain* that collects ALL logs from EVERYTHING (servers, agents, network sniffers, SNMP traps) and stores them in its database.	It provides the historical data and real-time context that the AI analyzes to find threats.
Alert (Correlated Alert)	This is an automatic, high-confidence warning generated when a **Correlation Rule** is triggered. *Example: Rule R-001 (Brute Force Success) is an Alert.*	Alerts skip the noise and tell you exactly where to look first. They are your highest priority tasks in the "Threat & Risk Center."
SOAR (Security Orchestration, Automation, and Response)	The *action layer*. This allows the AI Analyst (when given the right command) to tell the computer to perform security tasks instantly, like adding a firewall rule.	It drastically reduces the **"Time to Containment"** during an attack.

3. How to Talk to the AI Analyst (The Prompt Box)

The **Gemini AI Analyst** box is your direct command center. It intelligently routes your input to the correct service (Database Query, Deep Analysis, or SOAR Command).

3.1. Rules for Asking Questions (Prompts)

Your prompt's *intent* determines what the AI will do.

Prompt Type	Goal	How to Ask (Example)	AI's Primary Action
1. Query/Search	To retrieve raw or	"Show me failed	Converts request

(Find Logs)	formatted log data for specific criteria.	logon attempts from the last 24 hours."	into an SQLite query against the internal database.
2. Analysis/Summarize (Deep Dive)	To get an explanation, risk summary, or root cause analysis.	"What is the biggest risk right now?" or "Analyze the log pattern on Agent-101."	Sends the user prompt and relevant recent logs to the powerful Gemini LLM for interpretation and detailed summary.
3. SOAR/Action (Execute Command)	To take an immediate, automated security action on the host machine.	"Block the IP 192.168.1.50." or "Unblock 8.8.8.8."	Parses the command and executes a system shell script (netsh on Windows) if the server has Admin privileges.

4. Deep Breakdown of Key Use Cases

Use Case 4.1: The PQCM Alert (R-012) - Post-Quantum Crypto

Your system includes checks for weak, non-quantum-safe cryptography.

- **Trigger:** If the system sees logs indicating the use of vulnerable algorithms (e.g., SHA1, MD5, RSA/1024) during a crypto operation (Win Event IDs 5058, 5061), it raises alert **R-012**.
- **Action Required:** You must investigate the application or service using this weak crypto immediately, as it risks future data compromise by quantum computing attacks.

Use Case 4.2: UEBA Trust Score Drop (R-011)

The system tracks high-value users (Admin, FinanceMgr, etc.) for unusual activity patterns.

- **Trigger:** If a critical user performs many actions very quickly (e.g., 3 actions per second for several seconds), this suggests an automated process or account takeover, causing a **Trust Drop**.
- **Action Required:** This is critical. Follow the AI's Remediation Summary (e.g., **"IMMEDIATELY suspend user session"**).

5. Understanding AI Alert Context (CACP)

The CACP model provides real-time, structured context directly on the alert card for rapid

Triage.

Field	Deep Explanation	Triage Impact
Remediation Summary	This is the absolute minimum, first-step action. It is designed to achieve **Containment** (stopping the bleeding) immediately, before full investigation.	**T1 Focus:** Perform this step if SOAR cannot, or pass it to T2 to action.
Affected Systems Likelihood	A probability score (High/Medium/Low) predicting how likely it is that this single event has already spread to other systems in the network.	**Prioritization:** A 'High' likelihood means you should prioritize hunting for lateral movement immediately.
Predicted Next Action	The single most probable next step the attacker will take in the MITRE ATT&CK chain, based on the current attack stage.	**Proactive Defense:** You can use this to pre-emptively hunt for the predicted action, turning a reactive defense into a proactive one.

6. Manual Triage (T1 Role)

The T1 interface in the **Threat & Risk Center** is designed for a single function: **Validation and Escalation**.

- **Triage Controls:** There are no manual status updates (Pending, Dismissed) visible here. T1's job is binary: either the alert is not a security incident (in which case it shouldn't have fired, signaling a rule tuning need), or it is a valid threat that must be escalated.
- **T1 Action Flow (Escalate Only):**
  1. T1 reviews the alert card and CACP context.
  2. If the threat is real or requires deep dive: Click **Escalate to T2**.
  3. **Result:** A formal **Incident Ticket** is created, and the alert status changes to Escalated. T1 proceeds to the next alert.

7. Incident Handling (T2/T3 Role)

T2/T3 analysts manage the tickets in the **T2 Incident Ticket Queue**.

Ticket Status	T2/T3 Action	Linked Alert Management
OPEN (New Escalation)	Click **Investigate**. Status changes to IN_PROGRESS.	Alert remains visible to T1/T2 with Escalated status.
IN_PROGRESS	Follow the AI's investigation plan. Work is underway.	Alert remains visible with Escalated status.
RESOLVED/FALSE_POSITIVE	Select the appropriate resolution status (e.g., RESOLVED).	Ticket status is marked for final closure.
CLOSED (Final Action)	Click **Final Close** (after Report download).	**Crucial Synchronization:** The linked Correlated Alert is automatically marked as **'Solved'** in the database and immediately disappears from the Threat Center dashboard for ALL tiers.

8. Setup and Initialization Guide (Deep Breakdown)

This section explains in detail how to get the Guardian SIEM system running and the role of each component.

8.1. Prerequisites (What You Need)

Requirement	Why it is Needed	Action to Take
Python (3.8+)	This is the language the entire application (soc_dashboard.py, agent.py) is written in.	Ensure it is installed and added to your OS PATH.
Libraries (e.g., Flask, Scapy)	These packages provide core functions: Flask for the website, Scapy for network monitoring, google-generativeai for the AI, and reportlab for PDF	You must run the installation command exactly as written.

	generation.	
Administrator/Root Access	**CRITICAL:** For the SOAR actions (blocking IPs) and for agent.py to read system-level event logs (like Security logs), the script must have high privileges.	Right-click your terminal/command prompt and select "Run as Administrator."
Templates Folder	Flask must be able to find the HTML files (dashboard.html, login.html, etc.) to display the web interface.	Create a folder named templates/ in the same directory as soc_dashboard.py.

8.2. Component Roles and Startup Steps

You need to run up to three main components, ideally in separate terminal windows, to have a fully functional SIEM.

Step	**Component & File Name**	**Role and Importance**	**Command**
1. Install Dependencies	All Libraries	Gives Python the necessary tools (web framework, database connector, security tools) to run the SIEM.	pip install Flask Flask-Login werkzeug python-dotenv requests scapy google-generativeai pysnmp reportlab
2. Start the Agent (Optional but Recommended)	agent.py (Windows Event Log Collector)	**LOG SOURCE:** This script actively collects security, application, and system logs from your Windows machine and sends them to the SIEM server.	python agent.py

3. Start the SIEM Server (The Core)	soc_dashboard.py (Main Flask App)	**THE BRAIN:** Initializes the database (logs.db, users.db), starts all the background threads (Correlation Engine, DNS/GeoIP Enrichers), and serves the web dashboard.	python soc_dashboard.dashboard.py
4. Access the Dashboard	Web Browser	The User Interface (UI) where you monitor alerts, run AI commands, and manage tickets.	http://127.0.0.1:5000/
5. Login	User Management	Uses the users.db created during startup to authenticate you and assign your role (T1, T2, or T3).	Login screen (e.g., t1analyst / t1pass)

8.3. Troubleshooting Setup

Issue	Root Cause (What Went Wrong)	Deep Fix (Action to Take)
"ModuleNotFoundError"	A necessary Python library is missing.	Re-run the full pip install command to ensure all packages are present.
"404 Not Found"	Flask cannot find the requested URL route or the correct HTML file.	1. Ensure templates/ folder exists. 2. Ensure the URL you type exactly matches the @app.route() defined in soc_dashboard.py.
Server runs but logs	SQLite database file	Stop the server, **delete**

database fails	(logs.db or users.db) is corrupt or locked.	**logs.db and users.db files**, and restart the server (soc_dashboard.py will rebuild them).
SOAR/Agent permission fails	The script does not have Administrator/Root permissions.	Ensure you start both agent.py and soc_dashboard.py by using "Run as Administrator" in your terminal.